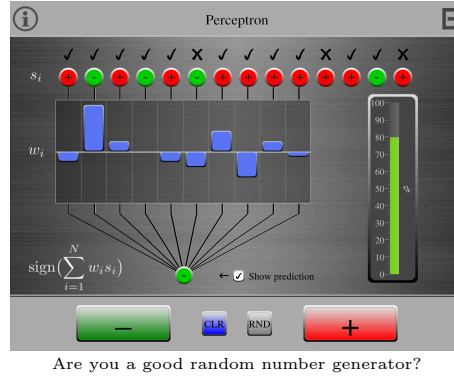


# PHYSICS OF COMPLEX SYSTEMS

LECTURE AND TUTORIALS – PROF. DR. HAYE HINRICHSSEN – MAXIMILIAN SCHÄFFERER – SS 2026



## EXERCISE 7.1: NAND AND XOR

(3P)

Let us represent Boolean values by FALSE=0 and TRUE=1. As activation function let us choose the step function

$$\sigma(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}.$$

- Find suitable weights  $\vec{w} = (w_1, w_2)$  and a bias  $b$  so that a *single* perceptron emulates a NAND gate. (2P)
- Show that it is impossible to realize a XOR gate with a single perceptron. (1P)

## EXERCISE 7.2: GRADIENT METHOD – STEEPEST DESCENT

(9P)

Since learning in artificial neural networks can be viewed as some kind of optimization problem, this exercise aims to make you familiar with the *method of steepest decent*.

Let us first consider a parabola  $f(x) = \frac{1}{2}ax^2 + bx + c$ . Choosing an arbitrary starting value  $x_0 \in \mathbb{R}$ , let us compute the iteration  $x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \dots$  through updates of the form

$$x_k \rightarrow x_{k+1} := x_k - \alpha f'(x_k),$$

where  $\alpha \in \mathbb{R}$  is a constant.

- Find the range for  $\alpha$  where  $\{x_k\}$  converges to the minimum of the function. (2P)
- Determine the optimal value for  $\alpha$  so that the iteration reaches the minimum already after a *single* iteration. (1P)
- Use this method to find the minimum of  $f(x) = \cosh(x)$ , starting with  $x_0 = 1$ . Choose the optimal  $\alpha$  and carry out the iteration numerically. (2P)
- Now let us be more general and consider a function defined on a vector space. To this end, let  $\mathbf{M}$  be a real symmetric  $n \times n$  matrix with nonzero eigenvalues and let the function  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  be given by

$$f(\vec{x}) = \frac{1}{2}(\vec{x} - \vec{a})^T \mathbf{M}^2 (\vec{x} - \vec{a}).$$

- Rewrite  $f(\vec{x})$  in the form  $f(\vec{x}) = \frac{1}{2}\vec{x}^T \mathbf{A}\vec{x} + \vec{b} \cdot \vec{x} + c$  and determine  $\mathbf{A}, \vec{b}, c$ . (1P)
- (e) Prove that  $f(\vec{x})$  has a unique global minimum at  $\vec{x}^* = \vec{a}$ . (1P)
- (f) The iteration rule has now to be generalized as follows:

$$\vec{x}_k \rightarrow \vec{x}_{k+1} := \vec{x}_k - \alpha \vec{\nabla} f(\vec{x}_k).$$

Compute the gradient  $\vec{\nabla} f(\vec{x}_k)$  and determine the optimal step size. (2P)

( $\Sigma = 12\text{P}$ )

Please submit your solution electronically as a single PDF file until Wednesday, June 03, 2026, via WueCampus.